

04_Integral Calculus

1. What is Integration?

Integration is the **reverse process of differentiation**.

- **Indefinite integral** → gives a **family of functions**
- **Definite integral** → gives a **number (area / accumulation)**

2. Indefinite Integrals

General form:

$$\int f(x) dx = F(x) + C$$

where

- $F'(x) = f(x)$
- C = constant of integration

Example

$$\int 2x dx = x^2 + C$$

3. Basic Integration Rules

(a) Power Rule

For $n \neq -1$:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Examples

$$\int x^3 dx = \frac{x^4}{4} + C$$
$$\int \sqrt{x} dx = \int x^{1/2} dx = \frac{2}{3} x^{3/2} + C$$

(b) Constant Rule

Example:

$$\int a dx = ax + C$$

$$\int 5 dx = 5x + C$$

Here,

$$\frac{d}{dx}(x + C) = 1$$

$$\int 1 dx = x + C$$

(c) Constant Multiple Rule

Example:

$$\int a \cdot f(x) dx = a \int f(x) dx$$

$$\int 3x^2 dx = 3 \int x^2 dx = x^3 + C$$

Here,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

(d) Sum and Difference Rule

Example:

$$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

$$\int (x^2 + 2x) dx = \int x^2 dx + \int 2x dx = \frac{x^3}{3} + x^2 + C$$

Here,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

(e) Exponential Rule

Example:

$$\int e^x dx = e^x + C$$

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$$\int e^{2x} dx = \frac{1}{2}e^{2x} + C$$

Here,

$$\frac{d}{dx}(e^x) = e^x$$

$$\int e^{ax} dx = \frac{1}{a}e^{ax} + C$$

(f) Logarithmic Rule

$$\int \frac{1}{x} dx = \ln|x| + C \quad (\text{for } x \neq 0)$$

Example:

$$\int \frac{1}{3x} dx = \frac{1}{3} \ln|x| + C$$

Because:

$$\frac{d}{dx} \left(\frac{1}{3} \ln|x| + C \right) = \frac{1}{3} \cdot \frac{1}{x} + 0 = \frac{1}{3x}$$